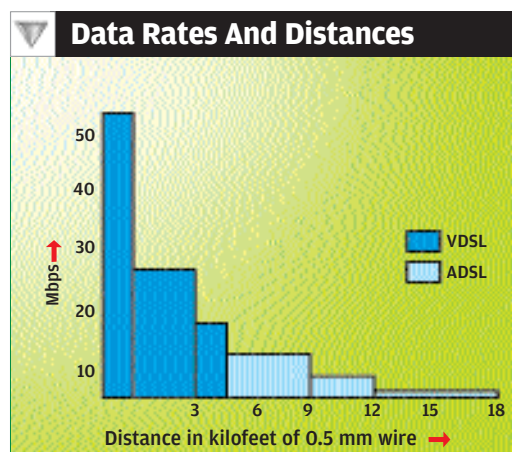




This graph is an indicative plot of connection speed versus distance from the service provider's CO. Because of the frequencies they use, the situation is quite different for ADSL and VDSL.

together to your service provider's Central Office (CO). At the CO, a *DSL Access Multiplexer* (DSLAM) aggregates all the ADSL lines coming in to it from all over the neighbourhood. Because of the frequency separation, the DSLAM can figure out what's voice and what's data. Voice is redirected to the PSTN (Public Switched Telephone Network), and Internet data is switched to the backbone connected to the ISP.

Now, the problem remains of the signal dropping as distance from the CO increases. See figure *Data Rates And Distances* for how much this drop is, with ADSL compared against VDSL. It's clear that ADSL can carry data reliably over much longer distances.

VDSL

Now here's the 54 Mbps we were talking about! VDSL offers such fibre-like speeds over copper wiring, but like we mentioned, some of the path from the home to the central office (full form: central telephone switching office) is indeed fibre.

VDSL isn't new. It's been around for a while, but is being rolled out mostly in Japan and Korea. We know it hurts, but all we can do right now is sit and watch providers in these countries battle it out to deliver the fastest speeds to their customers!

As indicators, in Korea, we're looking at 50 Mbps downstream and 11 Mbps upstream being offered. In Japan, it's more like 70 Mbps downstream and 30 Mbps upstream—courtesy VDSL 2 (See *Jargon Buster*). In Japan, NTT charges a premium of just about \$2 (Rs 90) per month over the cost of ADSL service!

So why doesn't everyone just move on to VDSL? The answer lies in what we've been talking about all along: fibre. (For example, markets in the United States have not been too responsive to VDSL.) To understand this, we need to understand what the last-mile problem is all about.

Fibre For The Last Mile

ISPs (In India, VSNL, Bharti and Reliance Infocomm) are connected to the Internet back-

bone. Telephone service providers (TSPs, such as MTNL) are connected to the ISP, and your home is connected to the TSP's exchange. In some cases, you might directly be connected to the ISP's central office (CO). If you're not connected directly to the ISP's CO, it's copper followed by fibre that carries data to the ISP.

It's the distance between the CO (or telephone exchange) and your home that is called the last mile, and here's where all the issues arise. "The last mile" typically means a couple of kilometres, but could be longer.

ADSL2+ (See *Jargon Buster*), within a mile (1.6 kilometres) of the CO, can deliver about 25 Mbps. That should be enough, one might say, but faster is better, and there's no questioning that. VDSL can reach 100 Mbps within 1,000 feet of the CO, but as the distance from the CO increases to more than a mile, the speed begins to compare with ADSL2+'s 25 Mbps.

Here's where fibre comes in. VDSL deployments use some length of fibre from the CO. How much, depends on the particular deployment. There's Fibre To The Neighbourhood (FTTN), Fibre To The Curb (FTTC), and Fibre To The Basement (FTTB). Of course, there's also Fibre To The Home (FTTH), which means fibre all the way with no telephone lines involved.

All these terms are somewhat self-explanatory, but to elaborate, FTTB is fibre all the way to the basement of a tall building, from where copper takes over to the individual offices in the building. FTTC means fibre would reach your street. And FTTN—the most practical possibility being talked about these days—involves the setting up of optical network units (ONUs) in a neighbourhood to which, on one side, fibre connects to the CO, and on the other side, copper connects to the home or office. The ONU's job is to convert the electronic signals from the copper to light signals, which are transmitted over the fibre.

Cable vs DSL

DSL, of course, isn't the only choice if you're looking at broadband in India—at present, cable is the most visible alternative. Whereas the infrastructure needed for DSL—namely, telephone wiring—already exists, each new cable Internet installation requires cables to be laid.

How do they compare? They're pretty neck-and-neck in terms of how good they are and what they can deliver. Both are capable of very high bandwidths; it's up to the service provider how much he wants to give you at an affordable price! About last-mile connectivity, for cable, it depends on how the equipment in your neighbourhood is wired up; for DSL, it depends on the quality of copper you're connected to. With cable, as the number of users on the network increase, your

speeds will go down; this is currently not the case with ADSL in India, where you're "guaranteed" dedicated bandwidths. Our guess, however, is that as ADSL penetration increases, the issue of sharing bandwidth will arise with ADSL as well, and you might not get dedicated bandwidth.

The situation, according to Sandeep Kolwadkar, deputy general manager (BD) at BSNL, is different for urban and rural areas: technicians in rural areas aren't proficient at setting up good cable networks, and that hampers speeds; in the case of DSL, it is poor copper quality that hampers speeds. Kolwadkar affirms that in urban areas, though, DSL does not suffer from the "poor copper" syndrome, and that connections are stable. That means we've only got to wait for prices to drop!